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标题	Synthesis of nano magnetic supported NHC-palladium and investigation of its applications as a catalyst in the Mizoroki-Heck cross-coupling reaction in H2O
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摘要（原文节选）

The unique properties of N-heterocyclic carbenes (NHCs) as an effective ligand for palladium catalytic centers have recently found much attention. In this work, Fe3O4 nanoparticle supported palladium(II) – N-heterocyclic carbene catalyst was successfully designed, synthesized and characterized by various methods. The activity of the catalyst was evaluated in the Mizoroki-Heck cross-coupling reaction in which the desired products were obtained in high yield in H2O as a green solvent. The reaction was carried out in sh ⋯

数据表 · Table 1（14 行，含数值行 14 行）

Optimization of the reaction conditions.a

Entry	Solvent	Base	Temp. (° C)	Catalyst (mol%)	Time (h)	Yieldb (%)
1	DMF	Et3N	90	0.272	7	90%
2	DMSO	Et3N	90	0.272	8	73%
3	CH3CN	Et3N	90	0.272	7	61%
4	EtOH	Et3N	90	0.272	9	80%
5	EtOH/H2O	Et3N	90	0.272	8	85%
6	H2O	Et3N	90	0.272	8	95%
7	DMF/H2O	Et3N	90	0.272	9	95%
8	H2O	K2CO3	90	0.272	9	78%
9	H2O	Cs2CO3	90	0.272	8	85%
10	H2O	Na2CO3	90	0.272	6	61%
11	H2O	Et3N	100	0.272	7	95%
12	H2O	Et3N	110	0.272	7	98%
13	H2O	Et3N	90	0.136	10	80%
14	H2O	Et3N	90	0.204	9	90%

脚注（逐字）：aReaction conditions: styrene, 4-iodotoluene, base (2 mmol), Fe3O4@Si – NHC – Pd and solvent (2 ml). bYields of isolated product.

正文对应句（逐字）：Modification of Fe3O4 NPs was confirmed by FT-IR spectroscopy (Figure 1-a) All samples showed a peak at 653 which is related to (Fe – O). The adsorption bands at 1082 and 821 cm − 1 are assigned to (Si – O – Si) bands, which are associated with Fe3O4 @SiO2. The broad adsorption at 3300 cm − 1 is due to – OH st

数据表 · Table 2（20 行，含数值行 20 行）

Heck reaction catalyst by Fe3O4@Si – NHC – Pda

Entry	R1	X	R2	Product	Time (h)	Yieldb (%)
1	4-Me	I	CO2Me	8a	8	95
2	H	I	CO2Et	8b	8.5	95
3	4-Me	I	Ph	8c	8	95
4	4-OMe	Br	CO2Me	8d	10.5	80
5	4-OMe	Br	CO2Et	8e	11	85
6	4-OMe	Br	Ph	8f	12	85
7	3-OMe	Br	CO2Me	8g	12.25	80
8	3-OMe	Br	CO2Et	8h	12.5	75
9	3-OMe	Br	Ph	8i	15	80
10	4-COMe	Cl	CO2Me	8j	22	75
11	4-COMe	Cl	CO2Et	8k	24	75
12	4-COMe	Cl	Ph	8l	24	75
13	H	Br	CO2Me	8m	10	85
14	4-CN	Br	CO2Et	8n	12	90
15	4-CN	Br	Ph	8 °	15	90

Entry	R1	X	R2	Product	Time (h)	Yieldb (%)
16	4-CHO	Br	CO2Me	8p	16	90
17	4-CHO	Br	CO2Et	8q	18	90
18	H	Br	Ph	8r	8	98
19	4-Me	I	CN	8s	8	90
20	4-COMe	Cl	CN	8t	8.5	80

脚注（逐字）：aReaction conditions: styrene, 4-iodotoluene, Et3N, Fe3O4@Si – NHC – Pd catalyst, in H2O at 90 ° C. bYields of isolated product.

正文对应句（逐字）：Modification of Fe3O4 NPs was confirmed by FT-IR spectroscopy (Figure 1-a) All samples showed a peak at 653 which is related to (Fe – O). The adsorption bands at 1082 and 821 cm − 1 are assigned to (Si – O – Si) bands, which are associated with Fe3O4 @SiO2. The broad adsorption at 3300 cm − 1 is due to – OH st

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